

PROMETHEUS Patent Description

Hybrid Analog–Digital Computing System with Time-Domain Multiplexed Computation

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1 DESCRIPTION

1.1 Field of the Invention

The present invention relates to computing systems and architectures, and more particularly to hybrid analog–digital computing systems implemented as programmable hardware accelerators. The invention further relates to time-domain multiplexed analog computation, calibration, and control methods suitable for high-speed, low-latency, and energy-efficient execution of continuous-time mathematical operations.

1.2 Background of the Invention

Digital computing systems dominate modern computation due to their flexibility, repeatability, and ease of programming. However, for certain classes of problems—particularly those involving

continuous-time dynamics, differential equations, optimization, control systems, and real-time physical modeling—digital systems face inherent limitations related to latency, power consumption, and discretization error.

Analog computing systems are capable of solving such problems by exploiting continuous physical phenomena. Historically, analog computers required extensive manual reconfiguration using physical patch cables and lacked programmability, scalability, and repeatability. These limitations resulted in analog computing being largely displaced by digital approaches.

Modern digital systems, including FPGAs, GPUs, and CPUs, frequently rely on sampling, quantization, and sequential arithmetic operations to emulate continuous-time behavior. Such approaches introduce latency, phase error, and energy overhead, particularly when solving short-horizon or real-time dynamic systems.

Accordingly, there exists a need for a programmable, scalable, and repeatable analog computing system that combines the benefits of analog computation with modern digital control, calibration, and software integration, while avoiding the limitations of manual analog reconfiguration.

1.3 Summary of the Invention

The present invention provides a hybrid analog–digital computing system implemented as a hardware accelerator, such as a PCI Express (PCIe) card, configured to perform time-domain multiplexed analog computation under digital control.

In one aspect, the invention comprises:

An analog computation fabric capable of executing continuous-time mathematical operations;

A digital control plane, implemented using programmable logic, configured to schedule, calibrate, and supervise analog computation;

A time-division multiplexing (TDM) mechanism that enables a finite analog resource to be shared across a plurality of logical computation channels;

A calibration and correction subsystem that compensates for analog non-idealities, drift, and variation;

A host interface enabling software-level configuration, control, and data exchange.

Unlike traditional digital solvers, the invention performs computation directly in the analog domain for defined time intervals, while using digital systems for orchestration, correction, and integration with software applications.

1.4 Brief Description of the Drawings

The invention may be better understood with reference to the following figures, which are illustrative and non-limiting:

Figure 1: System-level block diagram of a hybrid analog–digital computing accelerator showing host computer, PCIe interface, FPGA control plane, analog compute core, ADC/DAC banks, and calibration logic.

Figure 2: Internal architecture of the analog computing accelerator showing analog state elements, nonlinear elements, TDM switching network, and shared conversion resources.

Figure 3: Time-division multiplexed analog operation timing diagram illustrating discrete update instants and continuous-time evolution between TDM slots.

Figure 4: TDM coupling of multiple signals into a single analog state, demonstrating fan-in via time-division multiplexing and effective parallelism via temporal encoding.

Figure 5: Calibration and measurement feedback loop showing offset calibration, gain correction, and drift compensation pathways.

Figure 6: Software control interface illustrating host application, driver/API, PCIe interface, register map, and analog core control paths.

2 Detailed Description of the Invention

2.1 Overall Architecture

In one embodiment, the invention is implemented as a hardware accelerator card connectable to a host computing system via a high-speed interconnect such as PCIe. The accelerator comprises a digital processing subsystem and an analog computation subsystem.

The digital processing subsystem may include a field-programmable gate array (FPGA), system-on-module (SoM), or equivalent programmable logic device. The analog computation subsystem includes analog circuits capable of performing mathematical operations such as integration, summation, multiplication, and state accumulation.

The digital subsystem supervises the operation of the analog subsystem, including configuration, scheduling, calibration, and monitoring.

2.2 Analog Computation Fabric

The analog computation fabric comprises one or more analog computation elements arranged to implement continuous-time mathematical behavior. Such elements may include integrators, summers, multipliers, or other analog functional blocks.

The computation fabric is not statically wired. Instead, connectivity and functional assignment are controlled indirectly via digital scheduling and configuration, eliminating the need for manual patching.

The analog fabric may operate continuously or in discrete time windows as scheduled by the digital subsystem.

2.3 Time-Domain Multiplexing

To enable scalability, the invention employs time-domain multiplexing (TDM). In this approach, analog computation resources are shared among multiple logical computation channels by assigning each channel a time slot during which analog computation occurs.

Each time slot corresponds to a bounded interval during which the analog system settles and performs computation. Between slots, the configuration of the analog system may be changed under digital control.

Although computation is scheduled in discrete time slots, the computation within each slot is continuous-time and analog in nature, preserving the advantages of analog computation while enabling programmability and scaling.

2.4 Calibration and Error Compensation

Analog computation is subject to non-idealities including offset, gain error, temperature drift, and component mismatch. To address this, the invention includes a calibration subsystem.

Calibration may be performed:

At startup;

Periodically during operation;

On demand in response to detected error conditions.

Calibration data may be stored digitally and applied as correction factors during operation. The calibration subsystem may include analog-to-digital converters (ADCs), digital-to-analog converters (DACs), or other measurement and correction mechanisms.

2.5 Digital Control and Host Interface

The digital control subsystem manages communication with a host computing system. This may include:

DMA-based data transfer;

Register-mapped configuration;

Status and health reporting.

A software interface enables users to define computation parameters, schedules, and operating modes. The invention may be integrated into higher-level software systems, including workstation applications, control systems, or simulation environments.

2.6 Scalability and Variants

The invention is inherently scalable. In different embodiments:

The number of analog computation channels may vary;

The TDM schedule may be adapted to different timing requirements;

The digital control subsystem may be implemented using different programmable or fixed logic technologies.

The invention is not limited to PCIe form factors and may be implemented in other interconnect or embedded configurations.

3 Advantages of the Invention

The invention provides several advantages over prior art systems, including:

Reduced latency for short-horizon dynamic computation;

Improved energy efficiency compared to fully digital solvers;
Programmability without manual analog reconfiguration;
Scalability through time-domain multiplexing;
Compatibility with modern software ecosystems.

4 Industrial Applicability

The invention is applicable to a wide range of domains, including but not limited to:

Real-time control systems;
Hardware-in-the-loop simulation;
Optimization and numerical solvers;
Signal processing;
Scientific and engineering computation.

5 Scope of the Invention

The embodiments described herein are illustrative only. Variations, modifications, and alternative implementations will be apparent to those skilled in the art. The scope of the invention is defined solely by the appended patent claims.